Open Music Ontology

A minimal, non-authoritative bridge vocabulary that connects open cultural heritage and research metadata with common industry identifiers and release categories (version 0.1.0 - proof of concept)

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OMO: Open Music Ontology

The **Open Music Observatory (OMO)** does not seek to create new ontologies.
Instead, it **integrates and reuses existing, well-established data models** from the open data, research, and cultural heritage ecosystems.

In practice, this means that our data spaces and pilot projects (e.g. SKCMDb, Finno-Ugric Data Space, TextileBase) combine metadata derived from:

- **Open Government & Open Science:**
  DCAT-AP — the European data portal metadata model.
- **Cultural Heritage & GLAM:**
  EDM (Europeana Data Model),
  CIDOC CRM, and Records in Contexts (RiC).
- **Emerging Cultural Infrastructure:**
  High-Definition & Collaborative Cultural Data Infrastructure (HDTO).
- **Common Metadata Principles:**
  Dublin Core Terms (DCTERMS) — for general metadata interoperability.
- **Music Industry Standards:**
  ISWC for musical works,
  ISRC for sound recordings,
  DDEX for metadata exchange and release categorisation.

Our goal is not to add another layer of complexity, but to **connect these established models** so that cultural heritage data, rights management systems, and research metadata can interoperate seamlessly.

Semantic Architecture

The Observatory employs a **Wikibase-based ontology layer**, derived from the Wikibase Ontology.
This layer defines classes and properties that are compatible with the Wikidata ecosystem.

- **Equivalence links** (`owl:equivalentClass`, `owl:equivalentProperty`) map our entities to reference ontologies such as
`dcterms:` (Dublin Core), `rico:` (Records in Contexts), `crm:` (CIDOC CRM), `edm:` (Europeana Data Model), and `dcat:` (DCAT-AP).
- For **industry standards** like ISWC, ISRC, and DDEX that are not formally expressed in RDF/OWL,
the Observatory provides a minimal ontological scaffolding under the
`Open Music Ontology (omo)` namespace.
Source files are hosted at https://github.com/dataobservatory-eu/open-music-ontology.

This approach ensures that:

- Industry identifiers can be represented and queried alongside heritage and research data; and
- Semantic equivalence is maintained across research, rights, and cultural collections.

In short, **OMO acts as a semantic bridge rather than a new ontology.**

Summary

> The Open Music Ontology connects existing ontologies — it does not reinvent them.
> It provides a thin, open, and interoperable layer that unites public, research, and industry vocabularies to keep European music data FAIR and reusable.

Repository: https://github.com/dataobservatory-eu/open-music-ontology
Namespace (prefix): `omo:` → https://dataobservatory.eu/ontology/omo#
Status: Proof of Concept (v0.1.0)

Purpose

The **Open Music Ontology (OMO)** provides a **minimal, non-authoritative bridge vocabulary** connecting open cultural heritage and research metadata with **common industry identifiers and release categories** (e.g. ISRC, ISWC, album/single concepts).

It does **not** re-publish ISRC, ISWC, or DDEX standards.
Instead, it offers neutral linking terms (`rdfs:seeAlso`, `owl:sameAs`) that enable interoperability in knowledge graphs such as the Slovak Comprehensive Music Database (SKCMDb), Open Music Europe, and related Data Spaces.

Included Concepts (v0.1.0)

Type	Term	Description
Class	`omo:Recording`	A sound-recording entity (bridge for ISRC use).
Class	`omo:MusicalWork`	An abstract musical work (bridge for ISWC use).
Class	`omo:ContributorRole`	A role performed by a person or organisation.
Subclass	`omo:FeaturedPerformer`	A performer given primary credit.
Property	`omo:hasISRC`	Associates a Recording with an ISRC code.
Property	`omo:hasISWC`	Associates a Musical Work with an ISWC code.
Property	`omo:hasContributorRole`	Links an entity to its contributor role.
Property	`omo:agent`	Connects a role node to the performing agent.
Class	`omo:Release`	A curated bundle of recordings (album, single).
Class	`omo:ReleaseCategory`	Category describing a release type.
Subclass	`omo:Album`	A release consisting of multiple recordings.
Subclass	`omo:Single`	A release containing one main track.
Property	`omo:hasReleaseCategory`	Links a Release to its category.

Example (Conceptual)

```
turtle
@prefix omo: <https://dataobservatory.eu/ontology/omo#> .
@prefix ex:  <https://example.org/> .
@prefix foaf: <http://xmlns.com/foaf/0.1/> .

ex:rec001 a omo:Recording ;
  omo:hasISRC "SK0012345678" ;
  omo:hasContributorRole ex:role1 .

ex:work001 a omo:MusicalWork ;
  omo:hasISWC "T-123.456.789-Z" .

ex:role1 a omo:FeaturedPerformer ;
  omo:agent ex:johnDoe .

ex:johnDoe a foaf:Person ;
  foaf:name "John Doe" .
```

```
ex:release001 a omo:Release ;
  omo:hasReleaseCategory omo:Album ;
  omo:hasContributorRole ex:role1 .
```

Disclaimer

This ontology is not affiliated with IFPI (ISRC), CISAC (ISWC), DDEX, or any other rights holder. It simply provides neutral helper terms for research and cultural data interoperability.

For authoritative specifications, visit:

- ISRC: https://isrc.ifpi.org
- ISWC: https://www.cisac.org/Newsroom/news-releases/cisac-launches-major-project-upgrade-international-musical-work-identifier
- DDEX: https://kb.ddex.net

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